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**Hub Cultural: An Instagram-Mined, Human-Curated Pipeline for
Discovering Cultural Events in France**

From a Diaspora Network to a General Cultural Discovery Platform

Abstract

Culturally-relevant activity in the Paris region is rarely centralized: it is spread across hundreds of Instagram accounts and legible mainly to whoever already follows the right ones. This thesis builds a pipeline that discovers accounts through curated institutional seeds and a direct follower/following crawl, ingests their profiles and posts into a Neo4j graph, and detects events through a three-layer pipeline (semantic filtering, event-likelihood classification, LLM-based extraction and gating) validated by three rounds of blind human-versus-pipeline comparison, in which agreement rose from 58% to 96.7% and precision from 56% to 97.3% on a random sample — a targeted review of the twenty highest-scoring institutional events instead found twelve of twenty problematic, the slice a discovery product actually surfaces to visitors, and this thesis reports that gap rather than the headline figure alone. Because an automated account classifier selected a culturally valuable account under 10% of the time, hand curation of accounts replaces it as the trust signal the corpus relies on, with every design decision behind that pivot logged in real time so results remain traceable to the run that produced them. The pipeline feeds a public, bilingual, static discovery website with no backend, no user accounts and no personal data collection, publishing through a staged human review, a rolling retention window, and a deliberate manual deploy step reported here as method. An earlier component, an exact computation of account centrality (PageRank and betweenness), still contributes one term to the site’s ranking formula and one interface badge, but the account-selection role it was built for has been superseded by human curation. The thesis’s contribution is methodological (curation as an evidenced, not assumed, alternative to automated scoring; a blind evaluation protocol that measurably drove three pipeline redesigns; a transparent account of where automated signals fail silently), empirical (a categorised network and a growing, increasingly accurate catalogue of 751 events at the time of writing), and applied (a live, deployed tool usable by the community it describes).

Résumé

L'activité culturellement pertinente en région parisienne est rarement centralisée : elle se répartit sur des centaines de comptes Instagram et reste surtout visible à qui suit déjà les bons comptes. Ce mémoire construit un pipeline qui découvre des comptes à partir de sources institutionnelles sélectionnées et d'un parcours direct de leurs abonnements, collecte leurs profils et publications dans un graphe Neo4j, et détecte les événements à travers un pipeline à trois niveaux (filtrage sémantique, classification de probabilité d'événement, extraction et validation par grand modèle de langage), validé par trois cycles de comparaison humain-pipeline en aveugle, où l'accord passe de 58 % à 96,7 % et la précision de 56 % à 97,3 % sur un échantillon aléatoire — un examen ciblé des vingt événements institutionnels les mieux classés en a trouvé douze sur vingt problématiques, la tranche qu'un produit de découverte montre effectivement aux visiteurs, et ce mémoire rapporte cet écart plutôt que le seul chiffre principal. Un classificateur automatique de comptes ne sélectionnant un compte culturellement pertinent que dans moins de 10 % des cas, la curation manuelle des comptes le remplace comme signal de confiance sur lequel s'appuie le corpus, chaque décision de conception derrière ce changement étant journalisée en temps réel pour que les résultats restent traçables jusqu'à l'exécution qui les a produits. Le pipeline alimente un site web public, bilingue et statique de découverte d'événements, sans backend, sans comptes utilisateurs et sans collecte de données personnelles, publiant à travers une revue humaine par étapes, une fenêtre de rétention glissante, et une étape de déploiement manuelle délibérée, présentée ici comme faisant partie de la méthode. Une composante plus ancienne, un calcul exact de la centralité des comptes (PageRank et intermédiarité), contribue encore à un terme de la formule de classement du site et à un badge de l'interface, mais le rôle de sélection de comptes pour lequel elle avait été conçue a été remplacé par la curation humaine. La contribution de ce mémoire est méthodologique (la curation comme alternative documentée, et non supposée, au score automatique ; un protocole d'évaluation en aveugle qui a mesurablement motivé trois refontes du pipeline ; un constat transparent des cas où les signaux automatiques échouent silencieusement), empirique (un réseau catégorisé et un catalogue d'événements croissant et de plus en plus précis, 751 événements au moment de la rédaction), et appliquée (un outil en ligne, déployé et réellement utilisable par la communauté qu'il décrit).

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Chapter 1

Introduction

1.1 Context and Motivation

Hub Cultural¹ began as an attempt to make visible something that is well known anecdotally but poorly documented empirically: the cultural life that Latin American, and in particular Colombian, communities sustain in the Paris region through small associations, independent venues, cultural attachés, and a dense but scattered presence on Instagram. The author’s own Colombian background supplied both the initial motivation and the first seed account for data collection — the Colombian consulate in Paris (@consuladocolparis) — from which the network was grown outward through automatic relationship discovery.

As the project progressed, the nationality-based boundary that had organized it from the start — *is this account or event part of the Latin American diaspora?* — proved narrower than what the data actually supported: many of the best-connected cultural accounts discovered through the network turned out to be French institutions or mixed-nationality venues with no diaspora affiliation at all, but unmistakably part of the same cultural ecosystem the project set out to map. The operative criterion is now cultural relevance rather than nationality, with explicit priority — inherited from the project’s original four thematic agendas (docs/objetivos.md) — given to photography, cinema, literature, theatre, and a fifth, less formally bounded category: events organized by and for local communities, regardless of origin. This thesis treats that shift as a methodological finding rather than a footnote (§5.5): the nationality criterion turned out to be encoded not only in the project’s stated scope but, measurably, in the event-detection pipeline’s own reference vocabulary — a finding with a direct, quantifiable effect on what the system was able to see, detailed alongside the blind evaluation that surfaced it (§3.8.2).

¹Live discovery site: [hub-cultural-du.diegomerchanm.workers.dev](https://github.com/diegomerchanm/hub-cultural-du). Source code for the full pipeline described in this thesis: <https://github.com/diegomerchanm/hub-cultural-du>.

1.2 Problem Statement and Research Question

The cultural activity produced by, and adjacent to, migrant and diaspora communities in a host country is rarely centralized: it lives across dozens of Instagram accounts, is announced inconsistently, and is legible mainly to whoever already follows the right accounts. This is true of Latin American cultural life in the Paris region specifically, but — as this project’s own data eventually made clear — it is not a problem unique to any one national community; it is a structural feature of how small-scale cultural production circulates on social media generally.

This thesis asks: *How is culturally-relevant activity — with priority given to photography, cinema, literature, theatre, and local community cultural life — organized as a network of accounts and events in the Paris region, and can that structure be systematically mapped, analyzed, and made discoverable using publicly available Instagram data?*

This breaks into three sub-questions:

1. **Structural.** Who are the connective actors in this network — which accounts function as hubs or bridges between otherwise separate cultural circles, and does this differ between institutional accounts and individual or grassroots ones?
2. **Thematic.** How is cultural activity distributed across the project’s priority domains (photography, cinema, literature, theatre, community events), and how well does an automated pipeline classify it into these domains compared to human judgment?
3. **Practical.** Can the resulting graph be turned into a public-facing tool that helps the community it describes actually discover and attend these events — and what does building that tool reveal about the reliability of the underlying data?

1.3 Objectives

The project’s objectives, as originally framed (`docs/objetivos.md`) and updated here to reflect the current, cultural-relevance-based scope:

1. Build and maintain four thematic monitoring agendas — *literature*, *photography*, *cinema*, and *theatre* — tracking the organizations that play an active role in each, complemented by a fifth, less formally bounded priority: cultural events organized by and for local communities.
2. Consolidate a pipeline that extracts events from the accounts under these agendas, primarily from Instagram and, prospectively, from websites and other social networks.

3. Map the network of actors within each thematic agenda and categorize them — by type of organization, geographic zone, cultural identity where relevant, and structural role in the network.
4. Evaluate the participation opportunities these networks open up, using a deliberately human, non-technical success criterion: on this dimension, the project succeeds when it leads to actually meeting someone through it.

Geographically the project centers on Île-de-France, though a meaningful share of curated accounts and events originate from, or are organized by, institutions physically located outside France (§3.14). Instagram is the sole data source, a deliberate boundary rather than an oversight: it is the platform where this cultural activity is, in practice, announced, and only public accounts are in scope. Inclusion is governed by cultural relevance rather than national origin, with the four thematic pillars and the community-events category acting as explicit priorities rather than hard filters.

Chapter 2

Related Work and Positioning

This chapter positions the project briefly against the closest prior work rather than surveying a full theoretical apparatus. A scoping pass over comparable tools and a search of HAL and theses.fr found related work on Instagram data, cultural events and diaspora communication networks, but no project combining automated social-graph construction, NLP-based event detection and human curation into a live discovery tool for a migrant or cultural community in France.

The closest antecedents differ from this project on a specific point each. Dana Diminescu’s *e-Diasporas Atlas* (Diminescu, 2012; FMSH / e-Diasporas Atlas project, 2012) mapped migrant websites through hyperlinks; this project substitutes the Instagram @mention for that unit, a cheaper and more ephemeral signal. The Digital Methods Initiative’s *4CAT* (Peeters and Hagen, 2022; Digital Methods Initiative, 2022) captures data through manual browsing rather than automated scraping, trading scale for closeness to observation. Sciences Po médialab’s *Hyphe* (Jacomy et al., 2016) groups pages into researcher-defined entities, where this project keeps one node per Instagram account with no merging. *Osintgraph* (XD-MHLOO, nd) uses an almost identical technical stack for individual surveillance rather than cultural mapping, which is part of why this project’s ethical boundaries are stated explicitly (§3.13). Declarative directories such as the IOM/UN *Diaspora Mapping Toolkit* (International Organization for Migration, 2022) and *iDiaspora* (International Organization for Migration, nd) list actors who register themselves; this project is observational instead, surfacing accounts by detected activity — each approach is blind to what the other sees.

Chapter 3

Methodology

3.1 Research Design Overview

The project is organised as a seven-phase pipeline, each phase implemented as one or more independent, re-runnable scripts writing to a shared Neo4j graph, followed by the publication lifecycle described in §3.11. Figure 3.1 lays out all ten stages together with the script that implements each one. Script filenames are prefixed by pipeline phase (`1_harvest_`, `2_build_`, `3_analyze_`, `4_enrich_`, `5_export_`, DD-019), so a script's place in the pipeline is legible without opening it; the full source is public (§1.1).

Two design principles run through all seven phases and are treated in this thesis as methodological rather than merely technical. *Idempotency*: every script can be re-run without duplicating or corrupting what previous runs produced, because each checks a sentinel property before processing (`eventExtracted`, `lat`) or merges rather than inserts. *Separation of provenance*: the manual curation phase deliberately writes to a property namespace (`artType`, `culturalIdentity`, `geoZone`, `manualFollowersCount`) disjoint from the one the ingestion phase writes, so that neither can silently overwrite the other. The practical consequence is that automated and human-produced data remain distinguishable in the graph after the fact, which is what makes the curation-as-trust-signal argument of §3.8.1 verifiable rather than merely asserted. There is no scheduler or continuous integration: runs are manual and deliberate, since each execution costs money (Apify credits, LLM tokens) and the point of a run is usually to test a change rather than to keep a service fresh. Ingestion writes entirely with `MERGE` rather than `CREATE` and stamps every node with `firstSeenAt` on creation and `lastUpdatedAt` on every write, giving a per-node provenance trail without a separate audit table.

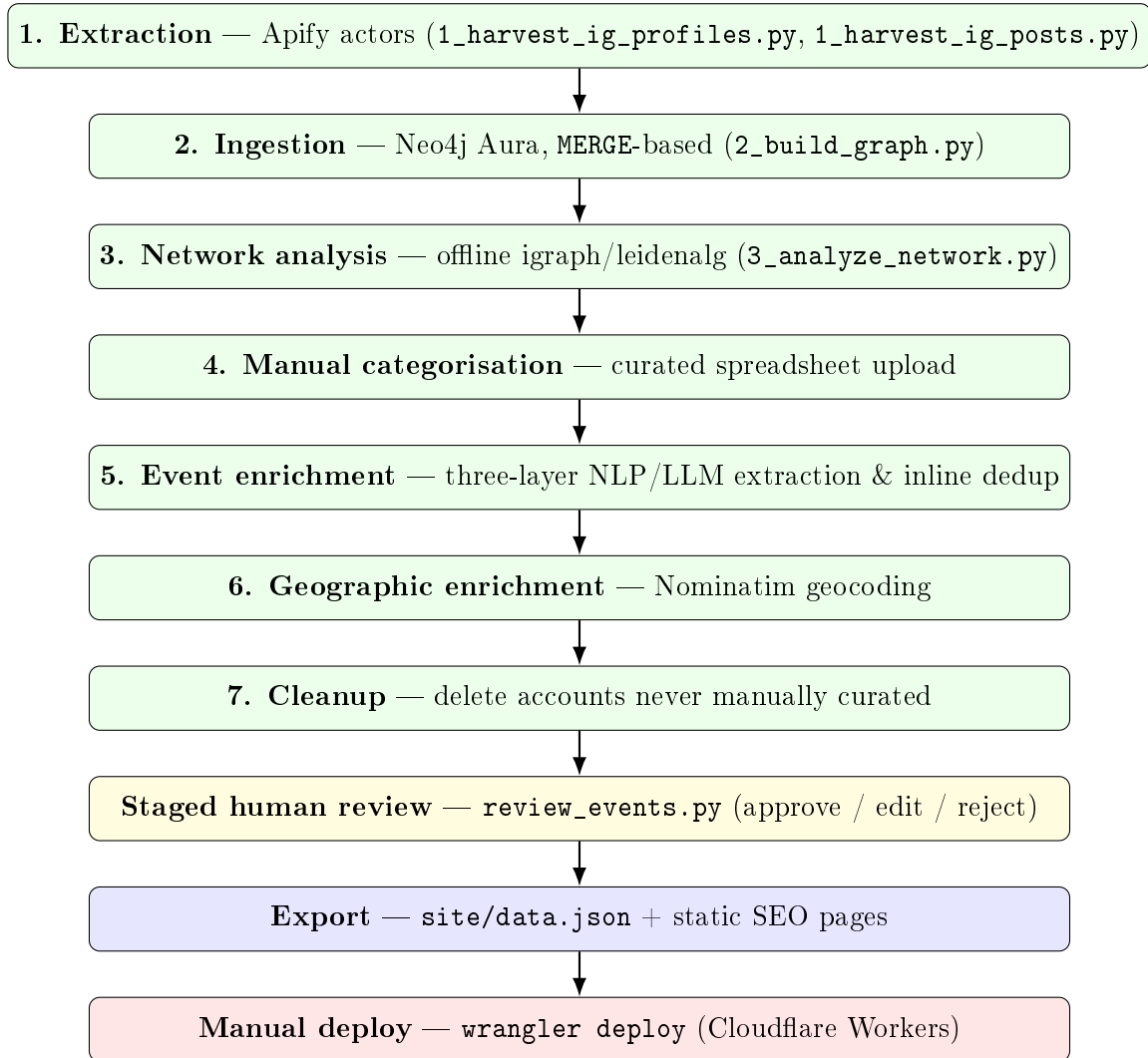


Figure 3.1: The pipeline’s seven ingestion and enrichment phases (§3.1), followed by the publication lifecycle (§3.11): staged human review, export, and a deliberately separate manual deploy step. Built from the architecture described in this chapter and `CLAUDE.md` rather than from `docs/data_flow.md`, which predates the three-layer LLM redesign and the review/export/deploy stages and was found, while building this figure, to be out of date.

3.2 Research Operations: Decision Logging, Run Tracking, and Iterative Safety

Independently of the technical pipeline itself, the project maintained three interlocking practices — a decision log, a run log, and dry-run-first design — that make its evolution auditable, which matters for an object of study that is a continuously growing graph, reprocessed repeatedly by successive versions of the extraction logic.

A real-time decision log. Every non-trivial design choice was recorded when made, in a sequentially numbered log (`docs/decisions_es.md`, running to DD-075 at the time of writing, with some numbers reserved but never used).¹ Each entry states the alternatives considered, the evidence behind the final choice and, where applicable, validation results (the deduplication resolver’s own validation run is reported in full where it belongs, §4.4, rather than repeated here). Recording decisions in real time, rather than reconstructing them afterwards, captures the alternatives considered and rejected *at the time*, which retrospective documentation loses.

A second document, `CLAUDE.md`, serves a different purpose: read by the AI coding assistant at the start of every session, it is deliberately overwritten whenever it goes stale, describing the pipeline’s current state rather than its history. Together the two files did most of the work of re-onboarding the assistant after each context reset — current architecture from `CLAUDE.md`, reasoning behind past choices from the decision log.

A run log. Every substantive execution of the pipeline against live data was logged with its date, parameters and results (`docs/runs_log_es.md`, RUN-001 to RUN-021), which made it possible to distinguish a genuine change in the data from a change in pipeline logic when comparing two exports, and to trace a decision back to the run that motivated it.

Dry-run-first design. Every script writing to the live graph ships with a `-dry-run` or `-diagnose` mode as a first-class feature — `cleanup_legacy_accounts.py` runs its deletion logic inside a transaction it rolls back under `-dry-run`, so preview counts are exact rather than estimated — which made it possible to iterate against a live, shared database without risking an irreversible mistake. This did not hold automatically, though: DD-038 records a production run that left almost no auditable trace because its diagnostic export was gated on the same flag; the fix restored parity between the two modes.

Together, these three practices make it possible to state, for any result in Chapter 4, exactly which version of the pipeline and which run produced it.

¹The log is maintained in Spanish, the language in which the project work was conducted; `docs/decisions.md` holds a partial English translation that currently stops at DD-033. Where entries are quoted in this thesis, the translation is the author’s.

3.3 Human-AI Collaboration in This Project

This project was built through a sustained collaboration between the author and an AI coding assistant (Claude, Anthropic): the author proposed a direction — a feature, a fix, a piece of analysis — usually without first working out its technical feasibility, and the assistant translated that into concrete, technically grounded options, stated their trade-offs, and implemented the one selected, so that the author interacted with the codebase almost exclusively at the level of intended behaviour while file edits, refactors and commits remained the assistant’s responsibility throughout. The same division of labour extended to the literature review, where an initial pool of roughly forty candidate references was assembled by the assistant and then verified and selected from by the author, rather than the two steps running in parallel. At the other end of the range sits the blind classification exercise behind the F1 figures of §3.8.2, the one instrument that could not be delegated at all, since judging whether a caption describes a genuine cultural event requires a familiarity with the community the assistant does not have. The pattern that holds regardless of how a given task was organised: work that can be fully specified — code, drafted text, summarised documentation — was substantially proposed by the assistant, while situated judgement — whether a detected event is culturally real, whether a candidate merge is correct — stayed with the author. The decision log of §3.2, which records alternatives considered rather than only the choice made, is what keeps the first category auditable after the fact.

3.4 Data Collection

Data collection relies on Apify’s hosted `instagram-profile-scraper` and `instagram-post-scraper` actors rather than a self-maintained browser-automation stack (DD-002): a hosted actor avoids the maintenance and IP-blocking risk of scraping Instagram directly, at a metered per-use price tracked in `.apify_cost_log.json` that constrains sampling decisions. Profile and post scraping are separate scripts (DD-003), since they use different actors, JSON structures and refresh frequencies.

Seed selection grew from the single seed account described in §1.1 (DD-006) into curated seed files (`config/seeds_*.json`) covering Latin American consulates and embassies in France plus connected French cultural and academic institutions (DD-022). Instagram’s own `relatedProfiles` suggestions add further candidates but are not treated as a trust signal: they enter a human-reviewed candidate pool (`discoveredViaCuratedAccount`, §3.8.1) rather than the corpus directly, and a later mechanism crawls the follower and following lists of trusted institutional seeds directly. DD-026 separates two questions the institutional seeds raise — whether an account is a valid cultural-analysis target, and whether it is structurally necessary to discover the

network — assigning consulates the role of seed source without treating them as cultural targets.

Post collection is recency-biased rather than density-maximising: the harvester computes a per-account window from the gap since its last known post, capped at a configurable maximum, merged into a rolling window of the fifty most recent posts (DD-028, DD-029), since the objective is detecting forthcoming events rather than reconstructing publication history.

3.5 Network Analysis: Status and Scope in This Thesis

Graph-theoretic analysis of the account network — exact PageRank and betweenness centrality, Leiden community detection, k-core decomposition, participation coefficient and E-I index — runs offline with `igraph` and `leidenalg` over two network layers, since Neo4j’s Graph Data Science library was unavailable in this environment (DD-013). An automated classifier built on top of these metrics was originally the intended criterion for deciding which accounts belonged in the corpus: §3.8.1 reports that under 10% of the accounts it selected turned out to be culturally valuable on manual review, so hand curation replaced it as the operative trust signal. Community detection, modularity and brokerage analysis are consequently not part of this project’s active methodology and are not reported as findings in Chapter 4; the full design, together with its one substantive result — some of the most structurally central accounts had almost no content of their own, direct evidence against a content-based selection criterion — is kept as closed work in §5.7.1.

One artefact of that earlier work remains live rather than archived: the account-level PageRank and betweenness values it computed are still read by the site’s ranking formula (§3.10, term *A*) and drive a single interface badge marking structurally bridging accounts. That residual dependency, and the caveat it carries — the values are not recomputed on an ongoing basis — is documented where it actually matters, alongside the formula itself, rather than here.

3.6 Event Detection Pipeline

Event detection runs in three layers over post captions.

Layer 1 — semantic candidate filtering. Each caption is embedded with `paraphrase-multilingual-MiniLM-L12-v2` and compared by cosine similarity against a reference set of roughly 100 phrases covering several event types, three languages and six announcement forms (calls, openings, listings, date-plus-place, registration, reminders). Embeddings rather than keywords were chosen because the corpus mixes Spanish, French and English with emoji and slang, where keyword matching fails on orthographic variation

and indirect phrasing (DD-007); the reference set is large rather than minimal (DD-010) because a narrow reference space misses exactly the small community events the project most wants to capture. Layer 1 exists primarily as a cost control: running the NLI model over every post was measured at five to seven hours on the available hardware, whereas pre-filtering reduces the candidate set to roughly 40% of the corpus and the total to about eighteen minutes (DD-009).

Layer 2 — is this event-like at all. Surviving candidates are passed to a binary natural-language-inference classifier (MoritzLaurer/multilingual-MiniLMv2-L6-mnli-xnli). Two decisions here are worth stating explicitly. The model is multilingual by requirement, not by preference: an earlier English-trained NLI model produced a detection rate of 0.7% on Spanish and French captions — effectively noise (DD-008). And the hypothesis sentence is now formulated in the caption’s own detected language, after a fixed-Spanish hypothesis was shown to depress scores for English and French captions regardless of their content.

Layer 3 — LLM extraction, typing and gating. Candidates that pass Layer 2 are sent to a large language model — Ollama locally, or one of four cloud providers tried in order with automatic failover and a cap on rate-limit waiting (Groq, Google, Cerebras, DeepSeek, DD-033) — which extracts title, description, date, location and price range in Spanish, together with French translations of the title and description generated in the same call at no extra request cost (`title_fr`, `description_fr`, DD-051, feeding the bilingual site of §3.12), assigns an event type from a fixed sixteen-label taxonomy, proposes one to three free-form artistic tags (DD-042), and — the decisive change — *gates* node creation. If the model reports that the post is not a public invitation, or not upcoming, no `:Event` node is created at all; in the earlier design the node was created regardless and merely scored lower, which §3.8.2 shows was the single largest source of false positives. If a provider is exhausted or fails, the pipeline falls back automatically across four cloud providers (Groq, Google, Cerebras, DeepSeek, in that order — DD-050) rather than stalling the run.

Three further constraints shape the Layer 3 stage. Calls are deduplicated by post identifier within a run, so co-published posts share one call. Reasoning traces are requested only in dry-run mode, since they are expensive and only useful for auditing. And a deterministic `has_text_date` gate — requiring that a date actually appear in the caption text — is evaluated *before* the model is called rather than after, which was measured to remove 40.7% of calls offline and 48.3% in production without changing a single final decision (DD-036). The date extractor behind that gate is itself constrained to avoid manufacturing precision: a month or year mentioned without a day is discarded rather than resolved to an arbitrary first-of-month (DD-034), after four disagreements in the evaluation sample were traced to exactly that behaviour.

Newly created events inherit `artType`, `institutionType`, `culturalIdentity`,

`geoZone` and `parentInstitution` directly from the publishing account’s manual curation, at no model cost, whenever that account has been curated. Deduplication happens twice: an inline resolver checks cosine similarity above 0.82 together with a location match and a date within three days before creating a new node rather than enriching an existing one, and a standalone resolver (§3.7) runs afterwards over the accumulated set.

3.7 Event Deduplication

Several accounts routinely publish about the same event, so deduplication is a structural feature of the corpus rather than an edge case. All pairs of events are compared by embedding similarity first, over a vectorised cosine matrix; dates function as mandatory confirmation rather than a primary filter, so where both dates exist they must fall within three days and similarity must exceed 0.75, and where one is missing a stricter threshold of 0.85 compensates for the lost corroboration. Location strings are not embedded, since text embeddings capture lexical rather than geographic similarity — two different theatres sharing the words "Théâtre de" would embed close together, while two addresses on the same street might not. Location instead backs a lightweight guardrail (DD-041): a merge is blocked only when both sides name a recognisable, non-overlapping country, drawn from a small gazetteer built from observed failure cases (generic community posts merging across cities that share vocabulary but not geography, such as Paris with Madrid). The guardrail is deliberately permissive — insufficient evidence on either side never blocks a merge — and its residual limit is documented rather than patched: it compares countries, not cities, so two different venues in the same city can still merge, an accepted trade-off recorded in RUN-021. The opposite failure — a missed merge — is also visible on the live site: an exhibition or a recurring event announced more than once, with shifting title or venue phrasing, can survive as separate cards, since a merge depends on title-embedding similarity and a three-day date window rather than on this guardrail, and neither is tuned for content announced repeatedly over a long run.

3.8 Deciding What to Trust: Curation and Blind Evaluation

Two questions govern what reaches the graph: which accounts belong in the corpus, and which of their posts are genuine events. The first is answered by hand curation, the second by a blind comparison between the pipeline’s decisions and the author’s own. They are reported together because the second is measured against the criterion the first encodes.

3.8.1 Human Curation as the Account-Level Trust Signal

The project’s original design selected accounts automatically: a classifier scored geographic and cultural similarity against reference embeddings plus hard profile signals, keeping accounts above a threshold. In practice, well under 10% of the accounts it selected turned out to be culturally valuable on manual review, so the pipeline’s authoritative signal shifted from that score to `manualDataCuratedAt` — written only by `load_manual_account_categorization.py` when an account appears in the hand-curated spreadsheet.

Three mechanisms explain the gap. The score rewarded visibility over cultural relevance, so well-documented commercial and institutional accounts outscored sparse small collectives. Its geographic component trusted a business-address field that was often wrong, patched first with a bounding-box check on latitude and longitude (DD-031) and then with a narrow exception for Alliance Française branches in Latin American cities, whose bios mention France as a subject rather than a location (DD-032). And structural importance is not content volume: the closed network analysis of §5.7.1 found top-bridging, top-PageRank accounts with almost no posts of their own, mentioned into relevance entirely by others — any criterion reading relevance off an account’s own output misses these.

Curation is reconciled with discovery — every Instagram `relatedProfiles` suggestion creates one account node on each scrape — by tagging rather than deleting: an uncurated account reachable via `RELATED_TO` from a curated one is marked `discoveredViaCuratedAccount` and kept as a review queue rather than entering the corpus. Human curation does not scale (§3.14), so the corpus’s boundary is set by how much one researcher can review.

3.8.2 Blind Human-versus-Pipeline Comparison

Event-level accuracy was assessed by classifying post captions blind — without access to the pipeline’s decision columns — and joining the labels against the diagnostic CSV the extractor emits. This was repeated at three increasing scales, each round feeding concrete fixes back into the code, as summarised in Table 3.1.

Table 3.1: Blind evaluation rounds and the code changes each one produced.

Round	N	Agreement	Precision	Recall (F1)
1 (2026-08-09)	100	58%	56% (90% gated)	0.865 (F1 0.682)
2 (through 2026-08-12)	495	—	82.4%	0.858 (F1 0.841)
3 (RUN-020, 2026-08-12)	150	96.7% (145/150)	97.3% (36/37)	0.90 (36/40) (F1 0.935)

Round 1’s asymmetry mattered more than its headline agreement figure: 29 of 35 false

positives already carried the Layer 3 model’s own verdict that the post was neither a public invitation nor upcoming, meaning the system had the information and simply did not act on it, since node creation was not yet conditioned on those flags. Simulating the gate raised precision from 56% to 90%, which is what turned gating from a scoring adjustment into a creation condition (§3.6). The same round produced the bias finding referred to in §1.1: the reference phrases of Layer 1 had been written entirely in the vocabulary of Colombian diaspora events, so the false-negative rate was 4.3% for Colombian and Latin American accounts against 12.9% for general and French accounts — real events in French and English discarded just below the similarity threshold. A criterion intended to capture cultural relevance was, at the implementation level, partly capturing national vocabulary instead.

Round 2’s three code changes (DD-036) were eighteen additional reference phrases in an institutional register, a deterministic date-presence gate, and an explicit prompt instruction excluding course registrations and admission calls. Round 3, the first run against the modified pipeline rather than a simulation, traced four of its five remaining disagreements to specific lines of code and fixed them the same day (DD-037): a theatre season written "26/27" misread as a date, and a caption-truncation window that discarded dates appearing later in long flyer-style captions. Figure 3.2 shows the underlying batch-level trajectory — five 100-post batches (RUN-015 through RUN-019, 495 posts in total) plus the separate, larger 150-post RUN-020 — compressed into the three rounds above — including a dip to 83.8% at RUN-019, traced to a new cluster of accounts (Alliance Française branches in Colombia) whose posts share an event-like structure with course-registration announcements. Read as a sequence, the trajectory shows detection accuracy accruing through successive rounds of exposure to real data, each surfacing an error pattern the next round’s fix then closed.

Two caveats belong with these numbers. The human labels are the author’s own, so they measure agreement with a single annotator’s criterion rather than with a consensus ground truth — a criterion that itself needed adjudicating at least once, when exhibitions announced with an explicit closing date were ruled to count as events even where the surrounding text is retrospective (DD-035), a ruling the pipeline’s own behaviour does not yet fully match. And agreement rose across the three rounds partly because the pipeline improved and partly because the samples differ in composition, so the figures track a trajectory rather than three measurements of the same quantity.

3.9 Geospatial Enrichment

Location strings extracted by the event pipeline are geocoded with Nominatim, the Open-StreetMap geocoder, chosen because it is free, requires no API key, covers both France and Colombia adequately, and imposes a rate limit (one request per second, implemented here

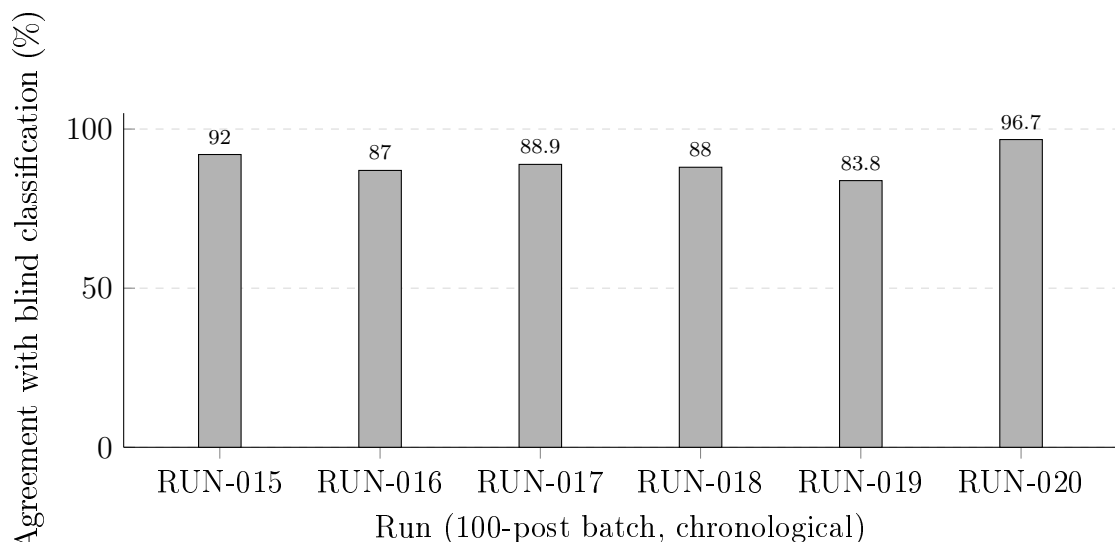


Figure 3.2: Batch-level agreement between the pipeline and blind human classification across the six batches underlying the "Round 2" and "Round 3" figures of §3.8.2 (docs/runs_log_es.md, RUN-015 to RUN-020): RUN-015, RUN-016, RUN-018 and RUN-019 are 100 posts each, RUN-017 combines two sub-batches of 49 and 50 posts (45/49 and 43/50 agreement, one co-published post collapsed as a duplicate) into a single 99-post bar, and RUN-020 — the first run against the modified pipeline rather than a simulation — is a separate, larger 150-post batch.

as 1.1 seconds between requests) that is tolerable at this corpus size (DD-018). Resolved locations are placed in an administrative hierarchy — **Location** → **Arrondissement** → **City** → **Country** — with Paris arrondissements derived from postal codes in the 750XX range. Sessions are logged so that a partial run can be resumed rather than repeated.

Roughly one event in seven never acquires coordinates, for two distinct reasons: some have no location text in the caption at all, and some have text that Nominatim cannot resolve. A city-level fallback was rejected as a display strategy because it produces coordinates that look precise and are wrong — an event placed at the centre of Paris because its address could not be parsed is indistinguishable, on a map, from an event that really is there. The discovery site (§3.12) therefore shows only events with genuinely resolved coordinates; this is a presentation filter, not a deletion, since the **:Event** node remains in the graph for analysis.

3.10 Ranking and Relevance Scoring

The discovery website ranks events with an explicit, documented formula rather than an opaque model, on the stated principle of contextual algorithmic curation rather than personalisation: there are no user accounts, and the same conditions produce the same ordering for every visitor. All component signals are converted to percentile ranks over the universe of publishable events before being combined. The score is

$$\text{Relevance} = 100 \cdot (0.30 Q + 0.22 A + 0.18 B + 0.20 T + 0.10 C) \cdot P$$

where Q is detection quality ($0.70 \cdot \text{eventScore} + 0.30 \cdot \text{confidence}$); A is the authority of the publishing account in the network, itself a weighted blend of PageRank (0.40), betweenness (0.25), follower count (0.20), participation coefficient (0.10) and verified status (0.05); B is social resonance (hotness score and post count); T is temporal proximity, implemented as explicit steps rather than an exponential decay so that it can be read and defended; and C is session context. P is a multiplicative penalty or bonus: political accounts $\times 0.55$, low-confidence extractions $\times 0.80$, free events $\times 1.05$.

Three aspects of this formula are methodological choices rather than tuning. First, follower count is deliberately the smallest term in A : a consulate with forty thousand followers is not thereby more culturally relevant than a collective with three thousand, and PageRank is weighted highest precisely because it measures recognition *within the cultural network* rather than general reach. Second, Q , A , B and P are precomputed at export time because they require percentile context over the whole dataset, whereas T and C are computed in the browser on each visit because they depend on the current date and on the individual session — precomputing them would freeze the ranking to the last export (DD-043, DD-044). Third, and discussed at length in §5.5, an earlier version of the formula included a $\times 1.05$ bonus for any event carrying a non-null `culturalIdentity` value; it was removed when the project’s scope broadened. A mandatory diversification pass then caps how many events from the same author, category or parent institution can appear together at the top of the home page, without which the largest category mechanically dominates the front page.

A caveat belongs here rather than only in §3.5: the PageRank and betweenness values behind A are not recomputed on a running basis. They are read off whichever account happened to be covered by the last executed pass of the offline network analysis, so an account curated afterward, or a shift in who mentions whom since that pass, is not reflected in A until the analysis is deliberately re-run. This is a live product dependency on a component this thesis otherwise treats as a closed, superseded line of work (§3.5); it is reported honestly here rather than smoothed over, precisely because it is the one place where that earlier work still has a measurable effect on what a visitor sees.

3.11 Publication Lifecycle and Event Retirement

Detection and ranking are only two stages of a longer cycle that this thesis treats as methodology rather than as an implementation footnote, because it is what makes the published catalogue trustworthy over time rather than only accurate at the moment of extraction. Three mechanisms close that loop, in the order an event actually passes

through them.

Staged human review. Every `:Event` node created by the extraction pipeline (§3.6) is tagged `:PendingReview` at the moment it is created, and both the JSON export that feeds the website and the Excel export used internally exclude that label by default. Concretely, this means the Layer 3 gate (§3.6) is a necessary but not sufficient condition for publication: passing it only qualifies an event to enter a review queue, not to appear on the site. A small local application (`review_events.py`) lets the researcher approve, edit or reject each pending event; rejection is deliberately soft — it adds a `:Rejected` label rather than deleting the node — so that no review decision is irreversible (DD-047). This reuses the same label-based status pattern already used for the automated gate rather than introducing a second status system for human decisions, and it is the direct methodological answer to §4.4’s finding that twelve of twenty highest-resonance institutional events were, on manual inspection, either not events at all or misclassified: automated gating alone was not sufficient to keep the published catalogue clean, and a second, human-executed gate was added downstream of it rather than tuning the first one further.

A rolling retention window, not deletion. Consistent with the project’s broader practice of never deleting research data outright (rejection above, and the researcher-only tier of the Excel export), `:Event` nodes are never removed from the graph once approved. Instead, retention is enforced only at export time: the export query excludes any event more than a fixed number of days in the past (thirty by default, configurable per run), and the website’s own client-side logic enforces the same cutoff a second time as a deliberate redundancy, so that a stale export cannot silently resurrect old events even if the cutoff at generation time were ever bypassed. The researcher-facing Excel export is exempt by design: it remains the full historical archive for analysis, while the public site only ever shows what is current. Before this cutoff existed, 71% of parseable events in the graph were already more than thirty days old and every one of them was still being shipped to every visitor; the fix is a filter on read, not a change to what is stored, which keeps the analytical record in Chapter 4 intact regardless of what the live site happens to be showing on a given day.

Publishing is a deliberate, separate step from generating data. Regenerating the exported JSON file, regenerating the static search-engine-facing pages built from it, and actually deploying the site to its hosting platform are three distinct manual commands, none of which triggers another. This was not the original design intent so much as a gap discovered in production: an approved batch of events did not appear on the live site because the data had been re-exported but never redeployed, and the site did not look broken, only stale, which is precisely why the gap went unnoticed for a period (DD-053). Collapsing the three steps into one command is listed as future work (§5.7).

3.12 The Discovery Website

The public artifact is live at `hub-cultural-du.diegomerchanm.workers.dev`. It is a static site: `5_export_dashboard_data.py` reads the graph and writes a single JSON file, and `site/` contains hand-written HTML, CSS and vanilla JavaScript that consumes it, deployed to Cloudflare Workers as a static-assets bundle (`site/wrangler.jsonc`, Workers Assets). There is no backend, no framework and no build step, since the publishable events fit comfortably in a file of a few hundred kilobytes and the site must be reconnectable from a plain repository without configuration (DD-043, DD-044).

Four design decisions are worth reporting as part of the thesis’s applied contribution. The category menu is fully dynamic — assembled on each load from the union of event categories and per-event artistic tags actually present in the data, with real counts ordered by volume — so it reflects what the upstream curation has prioritised rather than an assertion in the code. Session context is stored in `localStorage` rather than behind a login: category weights increment when a visitor filters by or opens an event in that category, decay daily, and are dropped below a floor, which yields the C component of the ranking formula with no server and no personal data. The export computes cosine similarity across all events and stores, per event, the identifiers of its five nearest neighbours, while the underlying 384-dimensional vectors are discarded and never reach the client. And the first version deliberately has no dedicated organiser-profile or methodology pages — those screens are more legible as thesis material than as product, and are listed as future work (§5.7) alongside Explore and Map.

Event content is bilingual only going forward, by design: interface text already switched between Spanish and French, but `title` and `description` were written once, in Spanish, by Layer 3. French versions are now generated by the same LLM call that extracts the event (DD-051) rather than backfilled onto the existing corpus, and the frontend falls back to the Spanish text whenever the French field is absent rather than leaving a blank card.

Screenshots of the home screen, category filter and event detail panel are not reproduced here — the live site (§1.1) is the more faithful record, and the interactions described above are easier to verify by visiting it than by reading a static capture.

3.13 Ethical Considerations

The corpus is restricted to public accounts and publicly visible posts; private accounts are labelled and not scraped. Organisational accounts — venues, institutions, collectives — are treated as public actors and named directly, since their posts are promotional by intent; individuals appearing incidentally in captions, comments or tags are not the object of analysis, and the cleanup phase removes accounts never touched by manual curation

together with their posts and comments, so this periphery is not retained indefinitely. The public site stores no personal data on any server — it is fully static, session preferences live in the browser’s `localStorage` with no third-party cookies, and event cards link back to the original Instagram post rather than reproducing it — and discloses that detection is automated, which matters since a language model sits in the pipeline and, as §3.14 documents, is wrong a measurable share of the time.

3.13.1 Legal Basis: GDPR and Instagram’s Terms of Service

Public accessibility does not remove personal data from GDPR’s scope, so this is addressed rather than assumed away. The basis relied on is legitimate interest (Article 6(1)(f)): a genuine, non-commercial academic purpose, necessary at this scale to answer a structural question about a whole network, balanced by the data having been manifestly made public by the accounts themselves and by the minimisation already in place (individuals are not the object of analysis, the uncurated periphery is deleted, §3.13). One field is a residual risk rather than a clean fit — `culturalIdentity` can encode an individual’s ethnic or national origin, and Article 9(2)(e)’s public-data exception holds less cleanly where it reflects the curator’s own reading rather than the subject’s own statement; it is optional, curator-observed, and never feeds an automated decision (§3.14). Individual notification (Article 14) is exempted for research under Article 14(5)(b) as disproportionate at this scale. Separately, scraping via Apify rather than Instagram’s own API contravenes Instagram’s Terms of Service — a known, accepted risk, not an oversight. This is a non-lawyer’s risk analysis, not a compliance certification.

3.14 Limitations of Method

The limitations below are drawn from failures actually observed in this project’s runs, not from a generic list of caveats.

Geocoding produces false precision when it fails. Around one event in seven has no usable coordinates, and the available fallback — resolving to a city-level hint — yields plausible-looking but wrong positions. The mitigation is exclusion from the map rather than approximation (§3.9), which trades coverage for honesty.

Date extraction is the pipeline’s most fragile component. Successive rounds of evaluation found dates invented from a bare month (DD-034), a theatre season "26/27" parsed as a calendar date, dates lost because the extractor truncated captions at 600 characters (DD-037), and implausible dates surviving a clamp that was silently skipped whenever either of two date parses threw — producing events dated 2036, 2052 and 2090 from text that mentioned an anniversary, an age limit and a duration in minutes (DD-039). Each was fixed once found; the pattern worth reporting is that all of them failed

silently, and all were found by manual inspection rather than by any automated check.

LLM classification is inconsistent at the margin. RUN-020 recorded two nearly identical posts about the same event receiving opposite verdicts, with no identifiable code cause. Layer 3 is also capable of typing an event with a label meaning "not a cultural event" while the surrounding logic created the node anyway, because creation did not consult the category field (DD-039).

Instagram-only sourcing bounds what can be observed. Organisations that announce primarily through mailing lists, their own websites or other platforms are absent by construction. The primary discovery mechanism (curated seed files and a direct follower/following crawl, §3.4) does not depend on Instagram's own recommendations, but the one secondary channel that does, `relatedProfiles`, is shaped by the platform's own visibility dynamics as to which further candidates it can surface at all — which is exactly why that channel feeds a human-reviewed candidate pool rather than the corpus directly. The algorithmic layer of the network is a proprietary artefact and is analysed separately for that reason (§3.5).

Curation does not scale, and the tier labels are sparse. The trust signal that replaced automated scoring is one researcher's judgement. The consequence is visible in the network analysis: at the time of RUN-015 only 62 of 7,269 account nodes carried an explicit `tier` label — the primary/secondary/excluded classification the project's original, since-retired account classifier assigned — so tier-based comparisons describe a small hand-curated subset rather than the graph as a whole, and this must be stated wherever such comparisons appear.

Geographic scope leaks through automatic discovery. Accounts outside the intended geography enter the graph through `relatedProfiles` before curation catches them — Alliance Française branches in Colombian cities being the recurring case (DD-032) — and at least one event in the institutional sample examined in §4.4 was a restaurant menu in Spain with no connection to the project at all.

A property inherited from the account is used as a proxy for the event's own location. Every `:Event` node inherits `geoZone` from the account that published it, at no additional extraction cost, and that inheritance has since become an operational filter: the human-review interface groups pending events by it for bulk triage (§3.11, DD-052). A Paris-based account can announce something happening elsewhere, however, so the property describes who posted, not necessarily where the event is. That the first bulk-filtered batch matched the reviewer's manual judgement in practice is evidence the heuristic works for the common case, not evidence that it is correct for every event it groups.

Chapter 4

Results

This chapter reports results from two sources: the event extraction and resolution of RUN-021 and subsequent runs, cross-checked where possible against the live export (`site/data.json`, 751 `:Event` nodes at the time of writing, prior to the retention filter of §3.11); and the evaluation trajectory of §3.8.2. Because the graph grows continuously, every figure below is stated with its date or run of origin rather than as a standing fact. A one-time structural snapshot of the account network — node and edge counts, community detection, bridging actors — was also produced (RUN-015, 2026-07-24) during the phase of the project described in §3.5, before the scope broadened and before the curation-based cleanup ran; it is not reported here as a current result, since neither precondition still holds, and is instead kept as a documented but closed finding in §5.7.1.

4.1 Anatomy of a Published Event

Before the aggregate figures below, it is worth being concrete about what a single `:Event` node actually contains, since no one stage of the pipeline produces it: a published event is the accumulated output of up to four separate stages, each writing a disjoint set of properties at a different point in the process, and confusing this with a single extraction step understates how much of the published record depends on work other than the language model call. Table 4.1 traces each group of fields to the stage that creates it; the full property-level schema, including `:Account` fields, is reproduced in Appendix A.

The practical consequence is that two events with identical titles and dates can differ in which of these fields they carry, purely because of when each was created relative to a given pipeline change: an event created before DD-051 has no French title regardless of how the site currently behaves, and an event whose publishing account was curated after the event itself was extracted inherits none of that account’s `artType` or `geoZone` until the extractor runs again over new posts from that account. What looks, on the published site, like one consistent object is therefore better understood as a record whose completeness reflects the pipeline’s own history as much as it reflects the event itself.

Table 4.1: Provenance of an `:Event` node’s fields, by pipeline stage.

Stage	Fields	Notes
Layer 3 LLM extraction (§3.6)	<code>title</code> , <code>description</code> , <code>type</code> , raw date, location text, <code>price_range</code> , <code>eventArtTags</code> , <code>eventScore</code> , <code>confidence</code>	Written once, at creation, only if the model’s invitation and upcoming flags both pass the gate; <code>eventScore</code> and <code>confidence</code> together form the Q term of the ranking formula (§3.10).
Same LLM call, no extra cost (DD-051)	<code>titleFr</code> , <code>descriptionFr</code> , <code>eventArtTagsFr</code>	Bilingual fields generated in the same request; never backfilled onto events created before DD-051 existed, so older events fall back to the Spanish fields on the site (§3.12).
Inherited from the publishing account, no model cost	<code>artType</code> , <code>institutionType</code> , <code>culturalIdentity</code> , <code>geoZone</code> , <code>parentInstitution</code> , <code>photoPermission</code>	Copied from the account’s manually curated fields (§3.8.1) only when that account has been curated; describes who published the event, not necessarily where or for whom it takes place.
Geocoding pass (§3.9)	link to a geocoded <code>:Location</code> node	Runs after creation, on a separate schedule; roughly half of events lack a resolved link at any given time (§4.3), for reasons that are diagnosed separately from whether the event itself is valid.
Deduplication (§3.7)	<code>hotnessScore</code>	The property the resolver compares when two records are judged duplicates, so that the canonical survivor is the one with more social resonance rather than simply the first one created.
Export time (§3.12)	identifiers of five nearest neighbours by cosine similarity	Computed once per export into <code>site/data.json</code> ; the underlying 384-dimensional embedding is discarded and never reaches this or any other export.
Review (§3.11)	<code>:PendingReview</code> / <code>:Rejected</code> labels	Added at creation and removed or replaced by a human decision in <code>review_events.py</code> ; an event with neither label has been explicitly approved.

4.2 Thematic and Event Landscape

At RUN-021 the graph held 664 `:Event` nodes, of which 559 met the validity criteria for publication and 170 passed the additional requirement of resolved coordinates and entered the 2026-08-14 site export alongside 200 accounts.

The distribution of the exported events by category was already uneven at RUN-021, with gastronomy the single largest category at 30% of the exported set — a pattern the current, larger catalogue confirms rather than corrects (Table 4.3, below).

Against the four declared thematic pillars, the picture is mixed. Table 4.2 shows the distribution of curated accounts and their events by artistic theme; it covers only the 114 accounts whose `artType` maps onto one of these eight pillar-aligned categories, not the full curated set — most curated accounts carry an `artType` outside this list entirely (gastronomy, institutional, community organising, and similar categories not itemised here), so the 114 is a deliberate subset rather than the denominator for the whole corpus.

Table 4.2: Curated accounts and their events by artistic theme (2026-08-13, derived from the curated `artType` field).

Theme	Accounts	Events
Visual arts	21	25
Music	23	24
Dance	21	20
Theatre	15	16
Literature	15	7
Circus	5	5
Photography	3	2
Cinema	11	2

A structural gap in the taxonomy compounds the detection numbers in Table 4.2: none of the sixteen event types available to the extractor corresponds to literature, so a book launch or a poetry reading has no correct label and falls into an adjacent category or none. Since literature is one of the project’s four founding pillars, this is a pipeline gap rather than an interface detail, and it motivated the addition of per-event artistic tags (DD-042) — which, being prospective only, do not yet cover the events already in the graph.

The catalogue has grown substantially since that snapshot. As of the current export (`site/data.json`, prior to the retention filter of §3.11), the graph holds 751 `:Event` nodes across 349 curated accounts, distributed as shown in Table 4.3. This 349 is measured differently from the 285 curated accounts cited in §5.5 (a 2026-08-31 snapshot restricted to `manualDataCuratedAt IS NOT NULL`): the site export’s account query is broader, also counting accounts that carry network-analysis metrics (`pageRankExact IS NOT NULL`) without manual curation, which is why it is the larger of the two figures rather than a

discrepancy. Only 5.6% of events in this snapshot are marked free, which is a property of what gets announced with a price attached in a caption, not necessarily of the underlying cultural field. Community events — the fifth, less formally bounded priority of §1.3 — have grown to 57 events, a large enough share (7.6%) to be worth tracking as its own line rather than folding it silently into the count.

Table 4.3: Event categories in the current export (751 events, 349 curated accounts).

Category	Events	Share
Gastronomy	210	28.0%
Institutional	132	17.6%
Visual arts	87	11.6%
Training and workshops	74	9.9%
Community events	57	7.6%
Music	44	5.9%
Audiovisual	44	5.9%
Performing arts	34	4.5%
Academic	29	3.9%
Festivals	27	3.6%
Politics	13	1.7%

A subsequent extraction pass populated `eventArtTags` across the catalogue, yielding 286 tag instances across 66 distinct free-text tags, led by Música (46), Artes visuales (33), Danza (28), Literatura (28), and Cine (25). Two results stand out against Table 4.2’s fixed-taxonomy counts. First, cinema and photography — the two founding pillars most under-detected there, with only 2 events each against 11 and 3 curated accounts respectively — register far more activity once tagged in free text: 25 mentions of Cine and 11 of Fotografía. This supports the earlier reading that the gap was a limitation of the fixed sixteen-label `:Event.type` taxonomy rather than an absence of real cinema and photography activity in the graph. Second, literature — which the fixed taxonomy could not detect at all, since none of its sixteen labels correspond to it — appears 28 times as `eventArtTags`, on par with dance, closing that gap through the same DD-042 mechanism rather than a change to the taxonomy itself.

4.3 Geographic Distribution

Of the 559 events meeting the validity criteria on 2026-08-13, 528 (94.5%) had an associated `:Location` node and 478 (85.5%) had real coordinates. The two shortfalls have different causes: 31 events carry no location text at all, because the caption never stated one, and 50 have location text that either failed to resolve in Nominatim or had not yet been through a geocoding pass. Only the first group is irrecoverable without a change to the extraction prompt.

These figures do not reconcile cleanly with the 170 events that reached the 2026-08-14 export (§4.2) or the 71 confirmed trustworthy the next day (§4.4) — not a bookkeeping error, but three snapshots of the same geocoding backlog in motion: 478 is a same-day graph query mid-geocoding; 170 is an export snapshot taken hours earlier, under a since-revised policy (DD-045, verified against git history) that then dropped an event entirely rather than just its map pin when geocoding looked unreliable; and 71 is that same batch after the next day’s promiscuous-coordinate diagnostic. Consistent with, not in tension with, the intermittent-failure pattern discussed above.

Against the larger, current catalogue of 751 events, the shortfall is wider rather than narrower: 91.7% carry some location text (a venue name or address, short of a full coordinate pair), but only 47.4% have resolved coordinates. The gap between those two figures — text present but unresolved — is the geocoding backlog rather than a caption-level absence, consistent with an intermittent Nominatim failure observed on a subsequent full run (zero coordinates resolved out of 276 attempted, cause unconfirmed) rather than with a change in how often captions state a location at all. Because the discovery site excludes ungeocoded events from anything geography-dependent rather than approximating their position (§3.9), this backlog directly determines how much of the catalogue is usable in a geography-first product, and geocoding coverage is consequently one of the more consequential, and cheaper to fix, current data-quality gaps.

A direct HTTP check against Nominatim, issued outside the pipeline with the same compliant custom `user_agent` the geocoder script sets, returned 200 and a valid, well-formed response for a routine query (`Paris, France`). This rules out a blanket API-level block or a rejected user agent as the cause of the zero-coordinate run reported above. The fault is therefore more likely intermittent or specific to particular queries — rate-limiting under the pipeline’s own request volume, malformed location strings for some events, or a transient outage during that specific run — rather than a persistent configuration error, though which of these applies has not yet been isolated. The arrondissement/commune breakdown that would substantiate the Île-de-France scope claim of §1.3 with actual numbers remains open until a geocoding pass with recovered coverage has been completed and requeried.

4.4 Data Quality Findings

The failure modes catalogued in §3.14 are reported here with the measurements attached to them.

A manual review of the twenty highest-scoring institutional events found a much worse rate than the aggregate figures of §3.8.2 suggest: roughly eight were genuine public events correctly categorised, seven were not events at all (interviews, podcasts, press releases, a private meeting) that had passed the public-invitation gate, and five were real events in

the wrong category. Twelve of twenty problematic is explained by how the sample was chosen: highest resonance within the institutional category is exactly where the pipeline's two known weaknesses concentrate — a leaky invitation gate, and a tendency to type an event by the publishing account rather than by the post's own content. Aggregate accuracy over a random sample and accuracy within a targeted high-visibility slice are different quantities, and a discovery product surfaces the second.

Deduplication performance is measurable on the same run. Over 664 events, the resolver evaluated 220,116 pairs in roughly twenty seconds, identified 68 duplicate pairs, and consolidated them through 189 relationship redirections, transitive chains included. Of the pairs it rejected, 12,375 failed on similarity, 165,018 on the date window, and 6 on the geographic-conflict guardrail — the last group being precisely the failure cases that motivated the guardrail (§3.7).

A live run on 2026-08-15 (DD-045) puts a number on the geocoding false precision: of 170 events that then carried some coordinate, 74 shared one of four "promiscuous" points produced by the discarded `city_hint_only` fallback tier — unrelated location strings (an Instagram handle, a theatre in Medellín, a Paris address) all resolving to the same place. Removing that tier, together with a coordinate-clustering filter flagging any point shared by three or more distinct location names, left 71 events with a trustworthy pin.

A live query against the current graph supplies a second figure, on date extraction: of 751 `:Event` nodes, 6 (0.8%) carry an `eventDate` outside a plausible 1950–2028 range, all through the mechanism DD-039 identified — a number embedded in the caption read as the date itself, for instance an event dated 1941 from a caption naming "Hotel Mandalay 1941" and one dated 2060 from a caption addressing people "de 18 a 35 años". This is a lower bound: a date that is wrong but still lands inside the plausible range does not surface here.

Chapter 5

Conclusion

This thesis set out to map, analyse and make discoverable culturally-relevant activity in the Paris region from public Instagram data, through the four objectives of §1.3.

5.1 Objective 1: Thematic Agendas and Trustworthy Accounts

The four founding agendas — literature, photography, cinema, theatre — plus community events, are maintained today through 349 curated accounts (§4.2). An automated classifier was tried first and selected a culturally valuable account under 10% of the time, so hand curation replaced it as the trust signal (§3.8.1) — the project’s clearest methodological contribution, arrived at by measuring the alternative rather than assuming it.

5.2 Objective 2: An Event-Extraction Pipeline

Three layers — semantic filtering, event-likelihood classification, LLM extraction and gating (§3.6) — produce the 751-event catalogue of §4.2. Three rounds of blind human-versus-pipeline comparison moved agreement from 58% to 96.7% and precision from 56% to 97.3% (§3.8.2). Photography and cinema looked under-detected against their curated account base, but the free-text `eventArtTags` data (25 Cine, 11 Fotografía mentions against 2 events each under the earlier count) points to a narrower measurement window rather than missing activity; literature has a separate, taxonomy-level gap instead — none of the sixteen event-type labels corresponds to it at all. Where the pipeline is not reliable (geocoding, date extraction, LLM classification, §3.14) is reported as part of the same result, not minimised.

5.3 Objective 3: Mapping the Network

Actor categorisation (organisation type, geographic zone, cultural identity) is answered by the same curated fields as Objective 1 (Appendix A). The structural half — who are the connective actors — was measured once, early, on a network since changed by scope and cleanup, and is reported as a closed, dated finding (92.5% individual accounts, a small institutional bridging core, some of the most central accounts carrying almost no content of their own) rather than revalidated here (§5.7.1).

5.4 Objective 4: Enabling Participation

This objective’s success criterion was deliberately human: leading to actually meeting someone. What exists is the infrastructure toward that end — a live, bilingual, static site with no backend, no accounts, no personal data collection, and a documented publication lifecycle (staged review, ranking, retention, a logged manual deploy; §3.12, §3.11). Whether it meets the human criterion remains untested: the site is live, but using it this way is a distinct act from building it, and that has not yet been attempted.

5.5 Scope and Its Cost

The project’s inclusion criterion shifted from nationality to cultural relevance in two places, one visible and one not until measured. The website’s ranking formula had applied a $\times 1.05$ bonus to any event with a non-null `culturalIdentity` — a field overwhelmingly Colombian and Latin American — favouring one set of identities by default; it was removed, and community detection was confirmed purely topological, unaffected by the field. Less visibly, the event detector’s reference vocabulary had been written entirely around Colombian diaspora events: the blind evaluation measured general and French accounts losing real events at 12.9% against 4.3% for Colombian and Latin American ones, on captions unambiguous to a human reader — the nationality criterion was encoded in the instrument, not only in the project’s stated scope. A 2026-08-31 snapshot of the 285 curated accounts confirms the population is now general rather than diaspora-specific (67.7% `culturalIdentity` = “No aplica”); the diaspora reading survives as one lens over a subset, not the corpus’s boundary.

The recurring lesson underneath all four objectives is that automated signals fail silently — a geocoder returning a precise, wrong coordinate rather than none, a language model typing an event “not cultural” while the surrounding code created it anyway (DD-039) — and were caught by a human reading output, not by the system flagging itself. Human curation was the more reliable signal, but does not scale and embeds one researcher’s judgement, which the evaluation protocol was measured against rather than

an external ground truth. The practical response was to put a deterministic, checkable gate in front of and behind every probabilistic component, and reserve relevance judgement for a person.

5.6 Limitations

The seven limitations catalogued in §3.14 amount to boundaries of a few kinds: single-platform sourcing makes organisations that announce elsewhere structurally invisible; the geography is concentrated but not clean, since institutions physically outside France enter through curation and automatic discovery alike; several pipeline components (geocoding, date extraction, LLM classification) fail silently rather than flagging low confidence; and the evaluation measures agreement with a single annotator’s judgement, not a consensus ground truth. That last point is also a cost: the corpus’s boundary is set by how much one researcher can review by hand, the direct price of replacing automated scoring with human curation (§3.8.1) — a trade this thesis argues was the right one on the evidence, but a real one. Every figure in this thesis is, for the same reason, a snapshot of a graph and a pipeline that were both still changing as it was written, stated with its run of origin rather than as a standing fact. To keep Chapter 4 reproducible against a fixed target rather than only a moving one, the export the 751-event figures are drawn from is archived, dated, in the repository (`data_processed/snapshots/data_2026-08-28.json`) rather than left implicit in a live database that keeps changing after submission.

5.7 Future Work

5.7.1 Revisiting Network Analysis on the Curated Graph

The account-network analysis described in §3.5 was run once, at RUN-015 (2026-07-24), against 7,269 `:Account` nodes, before the scope broadened beyond the Latin American diaspora and before `cleanup_legacy_accounts.py` removed the uncured periphery: a 92.5% individual-account graph around a small institutional core, dominated by a single giant component, with some of the highest- betweenness and highest-PageRank accounts carrying almost no posts of their own — the clearest evidence in the project’s own record for why an account-selection criterion should not read relevance off an account’s own content. That analysis is closed rather than current, since the criterion it served has been retired and the graph’s scope and composition have since changed; what remains open, and is left as future work, is whether the same pattern (a small institutional bridging core amid a large organic periphery) survives on the curated subgraph as it stands today, a smaller and cheaper question to ask than the original one, and worth asking on its own sociological merits rather than as a selection tool.

5.7.2 Pipeline and Product

Beyond the network-analysis question above, the project’s own planning notes and open items point to a concrete list: a new account classifier with explicit geography and cultural-relevance dimensions, should account discovery ever need to scale beyond what manual curation can review; an expanded seed set from Latin American consulates in Paris (DD-022 lists the handles, several still unverified); a revised harvest cycle prioritising cheap profile scraping before expensive post scraping; a literature label added to the event taxonomy and a backfill decision for `eventArtTags` (DD-042), directly motivated by the under-detection of photography and cinema events noted in §4.2; resolution of the open question in DD-035 on ongoing exhibitions; normalisation of stylised Unicode digits in captions if confirmed frequent (DD-036); the remaining site screens — a clickable arrondissement and commune map, an explore view, and organiser-profile pages; a backfill decision for `titleFr/descriptionFr` on events created before DD-051, together with a first quality check of the translations it produces; whether DeepSeek should remain in the default cloud cascade at all, given the empty-response failures of DD-050; and collapsing data export, static page generation and deploy (§3.11) into a single command so publishing stops being the step that gets forgotten (DD-053).

Appendix A

Graph Schema Reference

Table A.1: Node labels in the Hub Cultural graph.

Label	Origin	Notes
:Account	Ingestion	Also :Public or :Private; optionally :Political
:Post	Ingestion	Caption is the unit of event detection
:IgtvVideo	Ingestion	Treated like a post for authorship
:Comment	Ingestion	Author linked via WROTE
:Hashtag	Ingestion	
:Track	Ingestion	Audio track used in a post
:Location	Ingestion / events	Geocoded in phase 6
:Event	Event pipeline	Created only when the Layer 3 gate passes; also :PendingReview until a human approves it, or :Rejected if declined (soft reject, never deleted)
:Arrondissement	Geocoding	Derived from postal codes 750XX
:City	Geocoding	
:Country	Geocoding	

Table A.2: Relationship types.

Group	Types
Ingestion	PUBLISHED, HAS_HASHTAG, MENTIONS, TAGS_USER, COAUTHORED_BY, TAGGED_AT, USES_MUSIC, WROTE, ON, LOCATED_AT, RELATED_TO
NLP and geo	MENTIONS_EVENT, ORGANIZED, PARTICIPATED_IN, SUPPORTED, LOCATED_IN

Table A.3 gives a property-level view of :Account, grouped by which phase writes each property — the mechanism, described in §3.1, that keeps automated and human-produced data distinguishable in the graph after the fact. The equivalent view for :Event is Table 4.1 in the body (§4.1), which additionally notes what each group of fields is used for.

Table A.3: :Account properties by writing phase.

Group	Properties
Ingestion (2_build_graph.py)	id, fullName, biography, followersCount, followsCount, postsCount, verified, private, businessCategory, firstSeenAt, lastUpdatedAt
Curation (load_manual_account_categorization.py, COLUMN_MAP)	artType, institutionType, eventFrequency, parentInstitution, contentType, culturalIdentity, geoZone, manualFollowersCount, photoPermission, manualDataCuratedAt
Analysis (3_analyze_network.py)	pageRankExact, betweennessExact, wccId, kCore, participationCoef, leidenG05, leidenG10, leidenG15, tier

Appendix B

Selected Decision Log Entries

The full decision log (`docs/decisions_es.md`) is written in Spanish and runs to over 75 entries; it is the project’s real-time engineering diary, not a document written for this thesis. The entries below are the author’s own translations of the ones most load-bearing for the argument made in the body chapters, condensed for space — validation minutiae (exact test-case counts, `py_compile` confirmations) are omitted where they do not change the substance of the decision.

DD-006 — Seed strategy: BFS from a single institutional anchor (June 2026). The network was seeded from `@consuladocolparis` and expanded via mentions, tags and `relatedProfiles` rather than random scraping, on the reasoning that a consulate is the most central node of the Colombian diaspora in Paris — every relevant account sits within 1–3 degrees of it. A manually curated list of Colombian accounts in Paris was considered and rejected for researcher bias and for missing unexpected connections.

DD-026 — Separating seed role from cultural relevance. An earlier decision (DD-016) had put government accounts in a tier excluded from cultural-relevance analysis, since institutional accounts distort those rankings; but DD-006 and DD-022 use exactly those accounts (consulates, embassies) as the structural seed of the whole discovery process — without them there is no BFS expansion and no `relatedProfiles` to follow. The fix separates two questions that a single criterion had been conflating: whether an account is a valid cultural-relevance target (no, for consulates and embassies) and whether it should stay active as a discovery anchor (yes, unconditionally for this block) — implemented as `role="seed_source"` with `keep=False` regardless of computed score.

DD-031 / DD-032 — Geographic misclassification via a bounding box and an Alliance-Française-specific rule. A bio-text city-name fallback used to flag accounts outside France caught only two known cases; auditing raw `businessAddress` data found four more accounts with real `latitude/longitude` outside France but no diaspora-city text in their bio, invisible to the old fallback. The fix reads coordinates directly and penalises anything outside a France bounding box (`lat 41.0-51.5`, `lon`

-5.5-9.7), which generalises to any country without a maintained city list — at the accepted cost of excluding French overseas territories, a deliberate scope choice given the project's Île-de-France focus. One residual case, an Alliance Française branch in Medellín whose `businessAddress` had no coordinates at all and whose bio mentioned "Francia" as a subject rather than a location, needed a second, narrower rule keyed to the username pattern `alian[cz]a.{0,4}frances` combined with a non-French city token — deliberately not generalised to any diaspora account with a home-city name in its handle, which would have produced false positives among legitimate accounts like `medellin_en_paris`.

DD-034 / DD-037 / DD-039 — Three rounds of date-extraction bugs. DD-034 stopped the date parser from inventing a day-of-month when a caption named only a month, after a comparison against blind human labels found the pipeline assigning specific calendar days to captions that only said "this March." DD-037, the first live run of the retuned pipeline (150 posts, 96.7% agreement), traced its remaining disagreements to a theatre-season notation ("26/27") misread as a date and a 600-character caption-truncation window that discarded dates appearing later in long flyer-style captions — both fixed the same day. DD-039, auditing 90 events already written to production, found two further patterns: events tagged by the language model's own taxonomy as not a real cultural event were still created, because node-creation logic checked the invitation flags but never the category label itself; and a 1,095-day date-sanity clamp silently did nothing on three events whose dates should have been rejected, because a single `try/except` wrapped both the event date and the post timestamp together, so a parse failure on either one skipped the clamp entirely rather than falling back safely.

DD-050 — DeepSeek's silent empty-content failures reorder the provider cascade. A live run of 315 posts through the four-provider cloud fallback (Groq → Google → DeepSeek → Cerebras) found DeepSeek failing more often than the other three, but not with a rate-limit error: it returns HTTP 200 with an empty response body, which crashes the subsequent JSON parse with a generic error that gave no hint of the real cause. Since DeepSeek is also the only one of the four with no free tier, the cascade was reordered to `groq → google → cerebras → deepseek`, and the empty-body case is now detected explicitly rather than left to fail as a generic parse error.

DD-052 — Filtering pending review by geographic zone. Reviewing the first real batch of pending events by hand, the author found himself rejecting every single "Fuera de Francia" ("outside France") event one at a time. `review_events.py` gained a zone filter (inherited from the publishing account's own `geoZone`) and a bulk-reject action over whatever the filter currently shows — with an explicit, unresolved caveat left in the log: `geoZone` describes where the publishing account is based, not necessarily where the event itself takes place, so the bulk action is a useful heuristic for the common case that motivated it, not a per-event guarantee.

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